

CDS6314 DATA MINING

Trimester 1, 2024/2025

ASSIGNMENT (20%)

Examining the Impact of Aiming Proficiency and Geographical Region on First-Person Shooter (FPS) Player Performance

By

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1.Project Description

Valorant is a free-to-play multiplayer first-person shooter (FPS) game developed and published by Riot Games, the same company behind the hugely popular game League of Legends. It was officially released on June 2, 2020. Valorant is a team-based tactical shooter where players assume the roles of "agents," each with their own unique abilities and playstyles. Two teams, consisting of five players each, compete in rounds. There are various agents to choose from, each with distinct abilities divided into four categories: signature ability, two purchasable abilities (bought with in-game currency earned during matches), and an ultimate ability that charges over time or with kills.

The primary game mode involves one team attacking bomb sites while the other defends. In the main game mode, one team, known as the attackers, must plant a bomb (referred to as the Spike) at one of several designated bomb sites on the map, while the defending team tries to prevent the bomb from being planted or defuse it if it is planted. Players earn in-game currency by performing well in rounds, winning rounds, and completing objectives. This currency can be used to purchase weapons, shields, and abilities at the start of each round. Valorant emphasizes strategic team play, communication, and precise shooting. Each round requires coordination between teammates to execute strategies effectively and outsmart the opposing team. The game features several maps, each with its own layout, objectives, and strategic elements. These maps offer diverse environments and require players to adapt their tactics accordingly. Valorant offers a ranked mode where players can compete against others of similar skill levels to climb the ranks and earn rewards. Riot Games regularly updates Valorant with updated content, including agents, maps, and game modes. The game has also developed a thriving esports scene with professional tournaments and leagues.

In Valorant, players can compete with one another to gain a ranking rating. The better a player is against the general population, the higher their rank will be. There are currently nine different ranks in Valorant, those being Iron, Bronze, Silver, Gold, Platinum, Diamond, Ascendant, Immortal, and lastly Radiant; Radiant being the highest rank. The dataset covers a number of information about Valorant players within the ranks of Immortal and Radiant during Episode 4: Act 3 of the game’s ranked season. The domains covered by the dataset are:

1. Player Information
2. Performance Metrics
3. Match Outcome
4. Agent Selection
5. Weapon Performance

2 Data Mining Task

The primary data mining task is to analyze and uncover correlations between various performance metrics in first-person shooter (FPS) games, specifically:

1. The correlation between win rate, headshot percentage, and kill/death (K/D) ratio.
2. The correlation between geographical region and player performance metrics.

2.1 Motivation

The primary motivation behind this project is to identify and understand the key factors that influence player performance in FPS games. By analyzing correlations between win rate, headshot percentage, and K/D ratio, we can uncover insights into how these metrics interrelate and contribute to overall performance. Additionally, examining the impact of geographical **regions** on player performance can provide valuable information about potential regional disparities and inform strategies to address them.

2.2 Aims

1. Correlation Between Win Rate and K/D Ratio
   - Aim: To determine how strongly a player's ability to achieve a high K/D ratio (indicating offensive prowess) influences their win rate (indicating overall game success).
   - Expected Outcome: A positive correlation is expected, where players with higher K/D ratios also exhibit higher win rates.
2. Correlation Between Headshot Percentage and Player Performance:
   - Aim: To investigate if players who are more proficient in aiming (indicated by headshot percentage) perform better overall in terms of win rate and K/D ratio.
   - Expected Outcome: A positive correlation is anticipated, where higher headshot percentages are associated with higher win rates and K/D ratios.
3. Correlation Between Region and Player Performance:
   - Aim: To explore if there are significant differences in player performance based on geographical region, considering factors such as internet latency, regional play styles, and access to gaming resources.
   - Expected Outcome: Identifiable patterns that indicate how regional factors impact player performance metrics such as win rate, K/D ratio, and headshot percentage.

2.3 Expected Outcomes

1. Insights on K/D Ratio and Win Rate:
   - A clear understanding of how important the K/D ratio is in determining the overall success of a player in FPS games.
   - Identification of the threshold K/D ratio that correlates with higher win rates.

2. Influence of Aiming Proficiency:
    - Quantification of the impact of aiming skills (headshot percentage) on overall performance metrics.
    - Recommendations for players on improving aiming skills to boost performance.
3. Regional Performance Analysis:
    - Highlighted differences in player performance across different regions.
    - Insights into how regional factors such as connectivity, local gaming culture, and infrastructure influence FPS gameplay.

By executing this data mining task, the project aims to deliver actionable insights that can help improve player performance, game design, and competitive fairness in FPS gaming.

3. Related Works

3.1 Cultural influences on play style

The paper "Cultural Influences on Play Style" by Mateusz Bialas, Shoshannah Tekofsky, and Pieter Spronck explores the impact of national culture on how players engage with video games, specifically using data from the game Battlefield 3. The study aims to determine whether players from different nationalities exhibit distinct play styles and to what extent cultural differences influence these styles.

The study builds on Hofstede's cultural dimensions theory, focusing on three play style categories: competitiveness, cooperation, and tactical choices. The motivation for the research stems from the observation that video games often cater to specific cultural preferences, yet there is limited scientific research linking national culture to in-game behavior. The researchers used a large database (PsyOps) containing Battlefield 3 statistics, which provided extensive quantitative data on player behavior. The study employed ANOVA and MANOVA tests to analyze the data, seeking to identify significant differences in play styles among players from various nationalities.

Findings:

- Competitiveness: National culture accounts for 5.6% of the variance in competitive play style. Players from different countries showed significant differences in competitive behaviors.
- Cooperation: National culture explains 4.2% of the variance in cooperative play style. Notably, German and Swedish players exhibited more cooperative behavior than players from the UK and the US.
- Tactical Choices: The analysis showed that tactical choices were not significantly influenced by nationality.

The influence of national culture on play style, although statistically significant, is relatively small (around 5%). This suggests that while cultural tendencies exist, individual game situations and personal skills predominantly drive player decisions. The study confirms that players' behaviors align with cultural values associated with their nationalities, such as higher cooperation levels in more collectivist cultures like Germany and Sweden compared to the more individualistic UK and US. The findings suggest that understanding cultural influences on play styles can inform game design and development, potentially leading to more culturally tailored gaming experiences. The research highlights the importance of considering cultural dimensions in player modeling and game analytics. The study provides empirical evidence that national culture influences video game play styles, particularly in areas of competitiveness and cooperation. These insights contribute to the broader understanding of how cultural factors shape interactive media engagement and can

guide future research and game design practices to accommodate cultural diversity in gaming communities.

3.2 Investigating the Relation Between Playing Style and National Culture by Yaser Norouzzadeh Ravari , Lars Strijbos , and Pieter Spronck

In *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022)*, the study delves into the intricate interplay between nationality, cultural dimensions, and playing styles across four prominent Massively Multiplayer Online Games (MMOGs): Battlefield 4, Counter-Strike, Dota 2, and Destiny. To unravel these dynamics, the researchers meticulously curated datasets from diverse sources, including gaming platforms, community websites, and APIs, tailoring their selection criteria to ensure comprehensive coverage.

For Battlefield 4, player statistics were sourced from the BF4stats website via an API, capturing essential metrics like kill/death ratios, objective scores, and weapon usage for 14,906 players across 29 countries. Counter-Strike data, on the other hand, was obtained from the Steam Counter-Strike API, encompassing 13,962 players from 26 countries, with achievements categorized and weapon features grouped for analysis.

The Dota 2 dataset, procured from Opendota's API, honed in on active players within specific matchmaking rating brackets, yielding a rich tapestry of playing style features for 15,330 players spanning 30 countries. Destiny's dataset, sourced from DestinyTracker, spotlighted 8,240 players from 27 countries, capturing nuanced gameplay nuances such as kill-death ratios, weapon preferences, and teamwork dynamics.

Before diving into analyses, each dataset underwent meticulous preprocessing, including normalization and categorization of countries based on their Hofstede cultural dimension scores. With the groundwork laid, the researchers embarked on a multifaceted exploration, leveraging ANOVA tests, t-SNE dimension reduction, and machine learning models to unravel the intricate relationship between nationality, cultural dimensions, and playing styles.

Throughout their investigation, the researchers unearthed compelling insights, elucidating how players' cultural backgrounds intricately shape their gaming proclivities and behaviors. **Figures 3.2.1** and **3.2.2** visually elucidate the clustering of playing styles by nationality across different games, underscoring the profound influence of cultural factors.

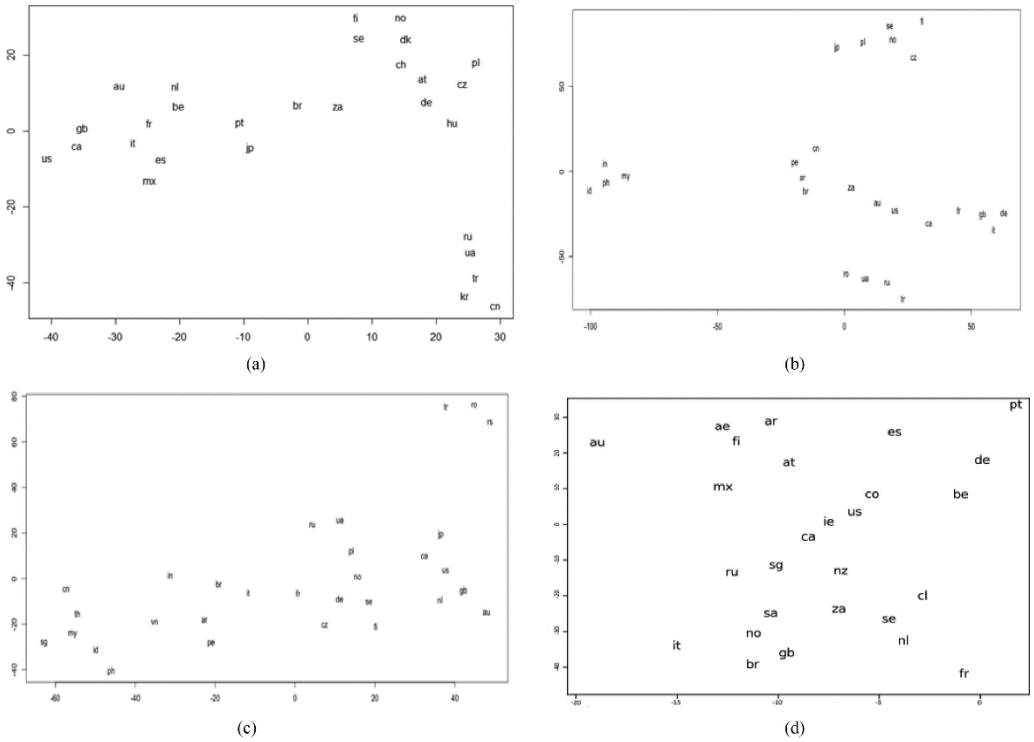


Fig. 3.2.1. Comparing playing styles and nationality in different games. (a) Battlefield 4. (b) Counter-Strike. (c) Dota 2. (d) Destiny. *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022).*

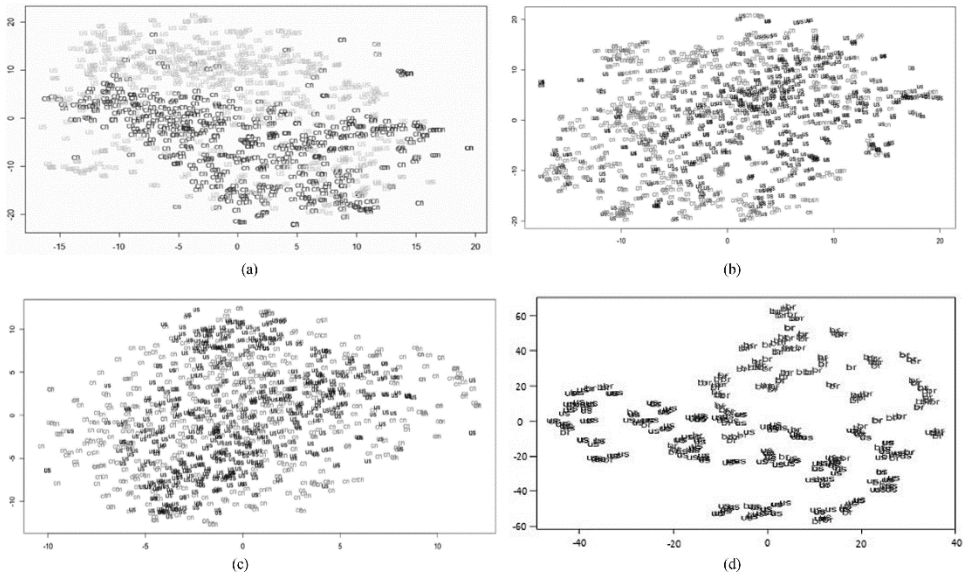


Fig. 3.2.2. Comparing playing styles in different video games. (a) China versus the United States in Battlefield 4. (b) China versus the United States in Counter-Strike.(c) China versus the United

States in Dota 2. (d) Brazil versus the United States in Destiny. *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022).*

Table 3.2.1. PR OPORTION VARIANCE EX PLAINEDBY EA CH HO FSTEDE DI MENSION

CATEGORYIN BATTLEFIELD 4 W ITHDF = (3,14902), *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022).*

feature	PD	IDV	MA	UA	LTO	IDL
Medals	.011	.037	.009	.002	.023	**.040**
Ribbons per round	.010	.035	.009	.003	.023	**.037**
Assignments	.006	**.015**	.003	.003	.010	**.015**
LM Gun shots fired	.014	**.058**	.004	.003	.023	.021
AR shots fired	.020	**.050**	.013	.010	.016	.028
SR shots fired	.011	**.034**	.012	.004	.016	.020
Shotgun shots fired	.011	**.028**	.010	.005	.010	.021
Suppression assists	.008	**.037**	.005	.000	.024	.020
Team score	.004	**.028**	.004	.000	.014	.012
Kill assists	.008	**.027**	.008	.003	.013	.015
Savior kills	.008	**.030**	.006	.003	.012	.014
Sum of variance	.111	**.379**	.083	.036	.184	.243

The bold values indicate the Hofstede dimension which explains most variance for each feature.

Table 3.2.2.PROPORTION VARIANCE EXPLAINED BY EACH HOFSTEDE

CATEGORY IN COUNTER-STRIKE WHERE DF = (3, 13 958), *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022).*

feature	PD	IDV	MA	UA	LTO	IDL
TR Map Nuke	.038	**.061**	.015	.025	.002	.014
TTA Ratio	.020	**.043**	.017	.018	.003	.006
TW Map Nuke	.031	**.050**	.013	.019	.002	.012
CSA Ratio	.023	**.043**	.016	.018	.001	.006
TR Map Dust2	.013	**.025**	.006	.008	.001	.011
TW Map Dust2	.012	**.023**	.006	.007	.001	.010
TR Map Inferno	.009	**.031**	.012	.015	.006	.008
TW Map Inferno	.009	**.030**	.012	.015	.006	.008
TW Pistolround	.004	**.015**	.011	.008	.006	.005
TR Map Train	.013	**.025**	.006	.011	.001	.011
TK Enemy Weapon	.002	**.013**	.012	.008	.003	.006
Weapons Donated	.001	.009	**.015**	.003	.004	.009
TK Enemy Blinded	.001	**.012**	.009	.006	.002	.006
TK Zoomed Sniper	.011	**.020**	.005	.006	<.001	.005
TW Map Train	.008	**.016**	.003	.006	<.001	.006
Sum of variance	.275	**.518**	.199	.230	.050	.135

(TTA: Team tactics achievements, CSA: Combat skills achievements, TK: Total kills, TR: Total rounds, TW: Total wins.)
The bold values indicate the Hofstede dimension which explains most variance for each feature.

Table 3.2.3.PROPORTION VARIANCE EXPLAINED BY EACH HOFSTEDE DIMENSION CATEGORY IN DOTA 2 WITH DF = (3, 15 356), *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022)*.

feature	PD	IDV	MA	UA	LTO	IDL
Total Matches	.017	.032	.009	**.021**	.004	.009
Ping	.008	**.028**	.004	.012	.014	.021
Total Words Said	.001	.009	.004	.006	**.027**	.034
Hero Damage	.009	**.015**	.004	.009	.002	.006
Deaths	.008	.013	.007	**.018**	.003	.013
Level	.006	**.007**	.004	.006	.002	.005
XP per Min Avg	.008	**.009**	.003	.006	.001	.005
Hero Ranged	.007	**.018**	.003	.004	.001	.004
Assists	.006	**.014**	.006	<.001	.005	.009
Purch. Force Staff	<.001	.007	.001	**.015**	<.001	.004
Hero Disabler	.004	**.011**	.002	.010	<.001	.002
Purchase Blink	<.001	.004	.002	**.008**	.001	.006
Tower Damage	.001	**.006**	.003	.005	.001	.003
Hero Support	.008	**.015**	.005	.007	.001	.004
Hero Jungler	.011	**.013**	.002	**.013**	.001	.003
Sum of variance	.122	**.253**	.080	.165	.075	.147

The bold values indicate the Hofstede dimension which explains most variance for each feature.

Table 3.2.4. PROPORTION VARIANCE EXPLAINED BY EACH HOFSTEDE DIMENSION CATEGORY IN DESTINY WITH DF = (3,10 477), *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022)*.

feature	PD	IDV	MA	UA	LTO	IDL
Kill Distance	.040	**.080**	.048	.026	.045	.072
Player DTR score	.074	**.081**	.059	.038	.008	.011
Life Span	.039	**.093**	.064	.023	.047	.064
Assists Per Game	.023	.032	.026	.028	.029	**.046**
Precision Kills	.053	**.104**	.062	.028	.018	.028
Assists	.048	**.103**	.060	.023	.019	.023
Ability Kills	.045	**.102**	.065	.024	.021	.022
Player kills	.056	**.070**	.045	.038	.006	.008
Orbs Dropped	.048	**.104**	.056	.025	.019	.027
Deaths	.038	**.097**	.065	.018	.024	.016
Player Games	**.052**	.050	.046	.037	.005	.004
Player Time Played	**.051**	.047	.042	.034	.005	.004
Revives Performed	.037	.057	**.063**	.023	.018	.011
Revives Received	.035	.055	**.061**	.023	.018	.009
Orbs Gathered	.036	**.090**	.045	.025	.016	.020
Sum of variance	.673	**1.167**	.805	.413	.297	.363

The bold values indicate the Hofstede dimension which explains most variance for each feature.

Table 3.2.5.PREDICTING NATIONALITY AND CULTURAL DIMENSION FROM PLAYING STYLES IN BATTLEFIELD 4, *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022)*.

target	baseline	RF	LR	SVM	NN
Nationality	3.48	15.49	**18.00**	17.45	16.79
PD	41.33	47.35	48.11	**48.62**	47.30
IDV	41.29	46.94	49.97	**5.34**	47.66
MA	44.87	47.32	**47.63**	47.56	46.42
UA	48.25	5.24	51.31	**51.93**	31.07
LTO	34.51	42.43	44.25	**44.46**	38.89
IDL	44.94	49.83	52.00	**52.72**	52.09

The bold values indicate models with higher accuracy.

Table 3.2.6. PREDICTING NATIONALITY AND CULTURAL DIMENSION FROM PLAYING STYLES IN COUNTER-STRIKE, *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022)*.

target	baseline	RF	LR	SVM	NN
Nationality	3.87	16.79	18.08	**19.51**	16.65
PD	38.45	49.77	49.80	**5.13**	48.98
IDV	38.49	48.30	47.58	**48.48**	48.37
MA	53.81	**55.71**	53.71	54.57	55.25
UA	42.28	**54.57**	53.42	53.85	54.53
LTO	42.36	44.36	**44.65**	44.11	43.61
IDL	42.32	**48.55**	46.72	48.37	47.19

The bold values indicate models with higher accuracy.

Table 3.2.7. PREDICTING NATIONALITY AND CULTURAL DIMENSION FROM PLAYING STYLES IN DOTA 2, *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022).*

target	baseline	RF	LR	SVM	NN
Nationality	3.35	15.63	17.48	**18.13**	16.70
PD	36.69	**48.27**	45.61	47.49	47.66
IDV	36.56	44.60	44.47	**45.21**	45.05
MA	43.29	**52.05**	48.86	49.15	49.54
UA	39.94	5.85	5.55	51.17	**51.33**
LTO	43.29	**47.40**	45.50	46.74	46.35
IDL	49.97	**55.66**	54.39	53.84	54.23

The bold values indicate models with higher accuracy.

Table 3.2.8. PREDICTING NATIONALITY AND CULTURAL DIMENSION FROM PLAYING STYLES IN DESTINY, *(Ravari, Y. N., Strijbos, L., & Spronck, P. 2022).*

target	baseline	RF	LR	SVM	NN
Nationality	3.70	2.37	**7.70**	6.65	4.63
PD	31.43	22.85	**34.44**	31.54	33.60
IDV	25.81	13.03	24.38	17.67	**28.26**
MA	35.90	48.92	**61.64**	61.02	54.60
UA	3.56	**5.24**	32.66	37.33	44.22
LTO	31.43	36.6	19.71	28.6	**39.43**
IDL	38.10	3.00	46.09	38.76	**51.42**

The bold values indicate models with higher accuracy.

Tables 3.2.1-3.2.4 offer granular insights into the proportion variance explained by each Hofstede dimension category across the four MMOGs, shedding light on the differential impact of cultural dimensions on playing styles. Meanwhile, **Tables 3.2.5-3.2.8** showcase the predictive prowess of

machine learning models in discerning nationality and cultural dimensions from playing styles, offering valuable glimpses into the underlying cultural dynamics at play.

Additionally, the paper examines the correlation between win rate and kill/death ratio, revealing a nuanced relationship that underscores the importance of both individual and team performance metrics. However, specific details on the correlation between region and player performance in Valorant are not provided within the scope of this analysis. As for factors that improve a player's performance, the study suggests that cultural background and playing style preferences play significant roles, though no direct causal factors are identified beyond these correlations.

In sum, the study's findings underscore the pivotal role of national cultures in shaping gaming preferences and behaviors, advocating for a nuanced approach to game design that embraces cultural diversity and fosters inclusivity across global audiences.

3.3 The relationship between the quantity of practice and in-game performance during practice with tournament performance in esports: An eight-week study

The study ("The Relationship between the Quantity of Practice and In-Game Performance during Practice with Tournament Performance in Esports: An Eight-Week Study," 2021) discusses the variables affecting player success in competitive gaming by examining the connection between the amount of practice and in-game performance during practice with tournament performance in esports. Although the paper's primary focus is on how practice and in-game performance affect tournament results, it also lays the groundwork for future research into related topics like the role of geography and aim skill in First-Person Shooter (FPS) player performance. Esports research has demonstrated that a player's ability to aim, as indicated by statistics like as kill/death ratio and headshot %, is a major factor in predicting their performance in first-person shooter games. Gaining insight into how these aiming abilities affect overall performance can help with player strategy and training techniques.

Furthermore, due to variables like latency, server location, and regional playstyles, it has been suggested that player location affects performance. Examining the effects of geographic location on player performance can provide a thorough understanding of the competitive dynamics and esports environment. Further investigation into the effects of aiming proficiency and geographic location on first-person shooter player performance can expand on the groundwork established by studies such as Pluss's research and improve our comprehension of the intricate relationship among personal abilities, contextual elements, and competitive achievement in esports.

3.4 Assessment of Human Expertise in First-Person Shooter Games by Ian Donovan et al.

In our attempt to find the factors that improve a player's performance in first person shooters, we've attempted to scope down the limitations or attributes that can be related to aiming. Therefore, to scope down the limitations or attributes that can be related to aiming, or as described by Ian Donovan; target-shooting, we'll be referring to the attributes explained by Ian Donovan et al, on a paper titled Assessment of Human Expertise in First-Person Shooter Games (2022).

The method of assessment that is used in "Assessment Of Human Expertise in First-Person Shooter Games"is by utilizing an aim-training program called AimLab. In AimLab, there are multiple scenarios that are supposed to resemble real in-game scenarios. The selected scenarios for the research's assessment are two scenarios, namely Gridshot and Sixshot. The following image is a screenshot taken by Ian Donovan et al. for the scenario called Gridshot.

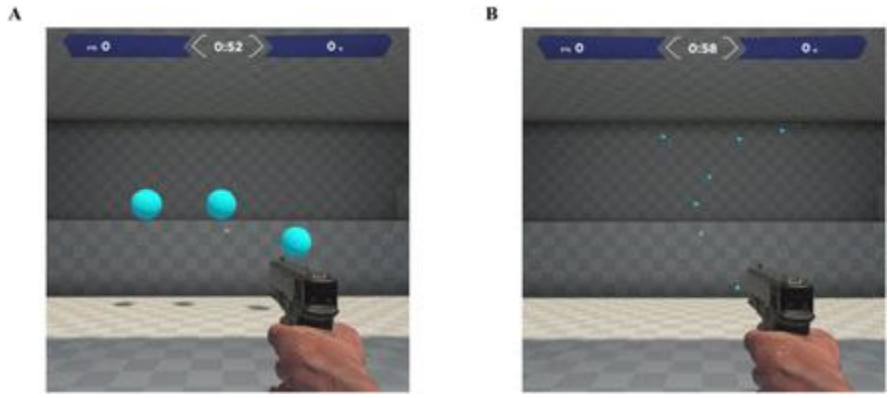


Figure 1 - Two of the selected AimLab scenarios; from left, Gridshot, Sixshot

A sample video representation for the respective scenarios can be found at shaman's (2021) and bardoz's (2021) channel.
An important component that Ian Donovan et al. were able to come out with during their research is the method to quantify an individual's motor capability, which they've abbreviated as SPAR, standing for "Speed, Precision, Accuracy and Reaction time, as well as swipiness". (Assessment Of Human Expertise in First-Person Shooter Games, 2021, Section 2.4.2 Movement Parsing). From their research paper, it is defined precisely on how these metrics are measured (2.4.3 Movement kinematics):

1. Speed (cm / second): peak speed at the midpoint of the movement.

2. Precision (1 / median absolute % distance to the target): variability in the accuracy across trials/targets.
3. Accuracy (% distance to the target): distance between the landing location of the movement and the center of the target.
4. Reaction time (sec): time interval between when the target appeared and the initiation of the movement, i.e., when the movement reached 5% of its endpoint.

3.4.1 Discussion

Through the findings gathered by Ian Donovan et al. on measuring a player's motor acuity, we've gathered that there are multiple methods we can follow to solely assess a player's target-shooting capability; but there is a problem in trying to correlate these findings to better do an evaluation based on the dataset we've selected. For example, to evaluate the speed movement of a player, Riot would need to implement a way to track a player's movement from the beginning of a game from the beginning of a match till the end; and there's also the problem of having downtimes during a real in-game match where it could affect the real evaluation of a player's target-acquisition speed. Evaluation of a player's reaction time in Valorant would also require Riot to implement a way for the game to recognize how fast a player was able to react seeing an enemy popup in the player's screen. Therefore, although the methods proposed by Ian Donovan et al. on the ways we can evaluate a player's target-shooting capability are all very elaborate and would likely be able to help a player to improve better, it is unfortunate that Valorant has no data that could help us analyze using the methods given.

4.Preliminary Task Distribution

No.	Name	Student ID	Task
1	LUQMAN BIN SAHADIN	1181202797	Data Mining Task: Related Work: Literature Review of Paper 3.3: The relationship between the quantity of practice and in-game performance during practice with tournament performance in esports: An eight-week study
2	CHRISTOPHER RANSIL PEERIS	1181203308	Related Works: Literature Review of Paper 3.2: Investigating the Relation Between Playing Style and National Culture by Yaser Norouzzadeh Ravari , Lars Strijbos , and Pieter Spronck
3	NABIL BIN HASANI	1181203473	Related Works: Literature Review of Cultural Influences on Play Style by Mateusz Bialas, Shoshannah Tekofsky and Pieter Spronck
4	PAUL JOHN C. ESCOBIA	1171103354	Project Description Literature Review of Paper 3.4: Assessment of Human Expertise in First-Person Shooter Games

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